

TASK 2 - ER MODEL AND RELATIONAL MODEL

DBMS PROJECT

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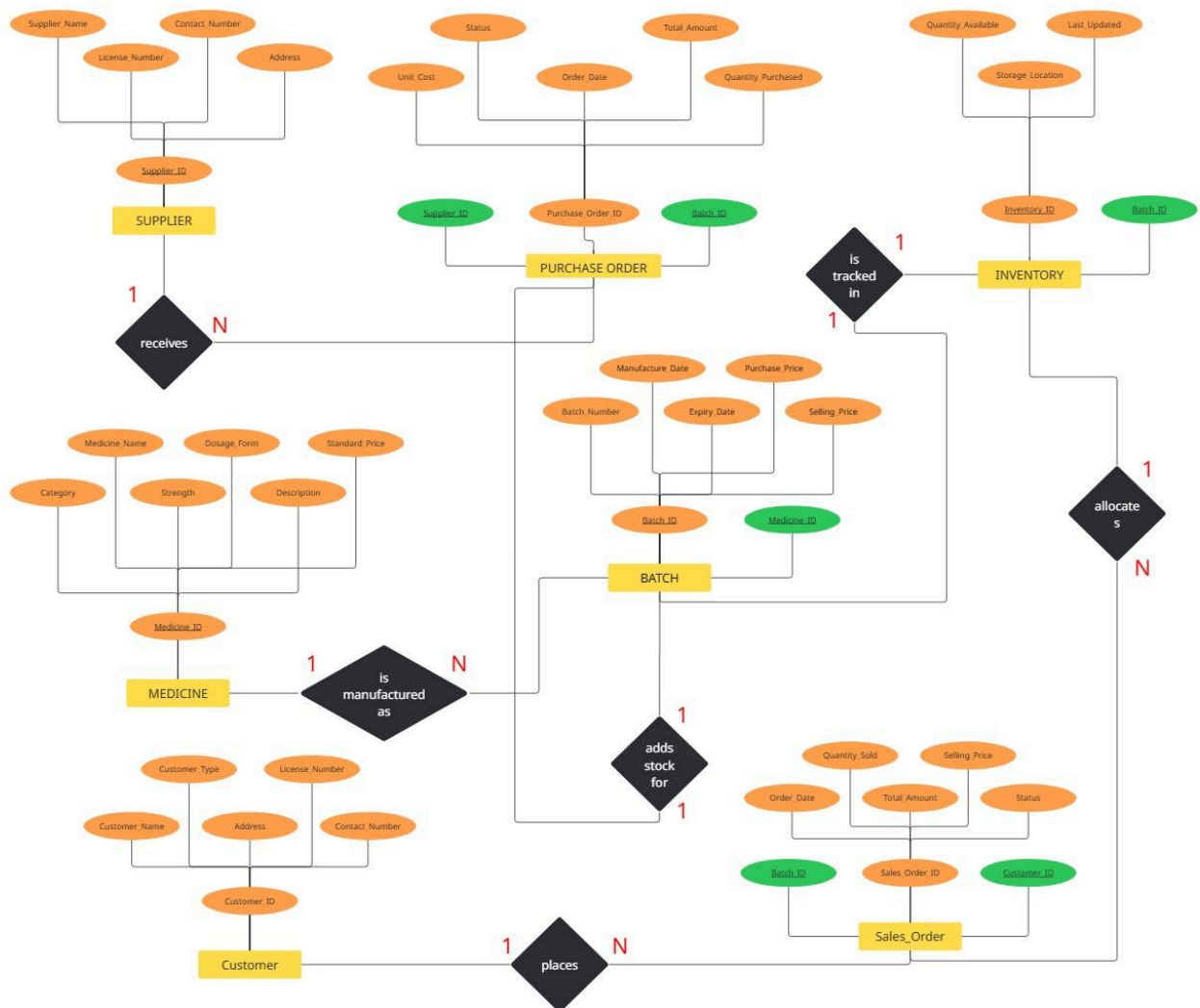
Project Group - 48

Tut Group - 4

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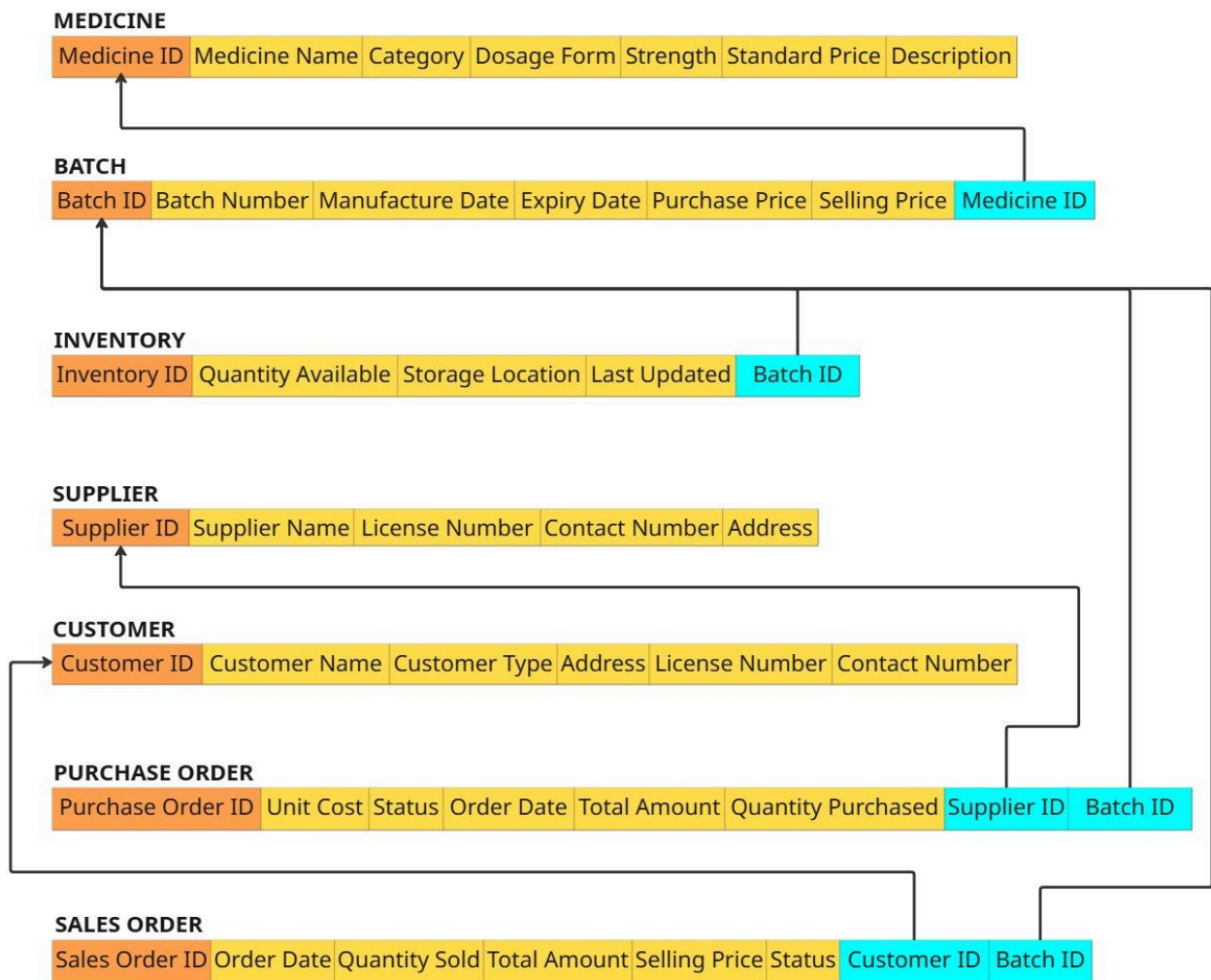
ER MODEL

ER MODEL



RELATIONAL MODEL

RELATIONAL MODEL



CONSTRAINTS OF ER AND RELATIONAL MODEL

Entity Constraints

- Each entity has a **unique primary key**.
- Primary keys are **mandatory (NOT NULL)**.

Relationship Constraints (ER Model)

1. Medicine – Batch

Relationship: *is manufactured as*

Cardinality: One-to-Many (1 : N)

- One medicine can have **multiple batches**.
- Each batch belongs to **exactly one medicine**.
- Medicine participation: **Partial**
- Batch participation: **Total**
- Min–Max:
 - Medicine: (0, N)
 - Batch: (1, 1)

2. Batch – Inventory

Relationship: *is tracked in*

Cardinality: One-to-One (1 : 1)

- Each batch is tracked by **exactly one inventory record**.
- Each inventory record refers to **only one batch**.
- Participation: **Total on both sides**
- Min–Max:
 - Batch: (1, 1)
 - Inventory: (1, 1)

3. Supplier – Purchase Order

Relationship: *receives*

Cardinality: One-to-Many (1 : N)

- One supplier can have **multiple purchase orders**.
- Each purchase order is linked to **exactly one supplier**.
- Supplier participation: **Partial**
- Purchase Order participation: **Total**
- Min–Max:
 - Supplier: (0, N)
 - Purchase Order: (1, 1)

4. Purchase Order – Batch

Relationship: *adds stock for*

Cardinality: **One-to-One (1 : 1)**

- Each purchase order is associated with **one batch**.
- A batch may exist **without a purchase order**.
- Purchase Order participation: **Total**
- Batch participation: **Partial**
- Min–Max:
 - Purchase Order: (1, 1)
 - Batch: (0, 1)

5. Customer – Sales Order

Relationship: *places*

Cardinality: **One-to-Many (1 : N)**

- One customer can place **multiple sales orders**.
- Each sales order is placed by **exactly one customer**.
- Customer participation: **Partial**
- Sales Order participation: **Total**
- Min–Max:
 - Customer: (0, N)
 - Sales Order: (1, 1)

6. Inventory – Sales Order

Relationship: *allocates stock for*

Cardinality: **One-to-Many (1 : N)**

- One inventory record (representing a batch) can fulfill **multiple sales orders**.
- Each sales order consumes stock from **exactly one inventory record**.
- Inventory participation: **Partial**
- Sales Order participation: **Total**
- Min–Max:
 - Inventory: (0, N)

- Sales Order: (1, 1)

Relational Schema Constraints

Foreign Keys(All foreign keys are NOT NULL)

- Batch → Medicine
- Inventory → Batch
- Purchase Order → Supplier, Batch
- Sales Order → Customer, Batch

UNIQUE Constraints

- Supplier License Number
- Customer License Number
- Batch Number
- Inventory.Batch_ID (ensures 1–1 Batch–Inventory)

CHECK Constraints

- Quantities ≥ 0
- Prices > 0
- Expiry Date $>$ Manufacture Date
- Order Date $\leq$ Current Date

Business Rules (Enforced via Triggers)

- Expired batches **cannot be sold**.
- Inventory **cannot go negative**.
- Inventory updates automatically after purchase and sales orders.
- Total Amount is computed from price $\times$ quantity.